# Department of CSE(AIML), PES University
# UE26CS151A -Python for Computational Problem Solving
# Laboratory Week 6

**Note:** Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

**Q1. Count the Number of Characters in a Text File** Write a program to read a text file and count the total number of characters present in it, excluding newline characters.

- **User Input:** File name entered by the user.
- **Data Types:** Use string for text, int for character count.

**Q2. Search for a Student in a Dictionary** Write a program to store student roll numbers and names in a dictionary. Read a roll number from the user and display the corresponding student name if it exists.

- **User Input:** Roll number (int) entered by the user.
- **Data Types:** Use dict with int keys and string values.

**Q3. Display Employees with Salaries Above a Given Amount from a CSV File** Write a program to read employee names and salaries from **week_6_set2_q3.csv** and display the employees whose salaries are greater than a user-entered amount.

- **User Input:** Salary threshold (int) entered by the user.
- **Data Types:** Use string for employee names, int for salaries.
- **File Reference:** Refer to **week_6_set2_q3.csv** for input.

**Q4. Count the Occurrences of Each Character in a Text File** Write a program to read a text file and use a dictionary to count the occurrences of each character, excluding spaces and newline characters.

- **User Input:** File name entered by the user.
- **Data Types:** Use dict with string keys (characters) and int values (counts).

**Q5. Store and Display Product Prices Using a Dictionary** Write a program to store product names and their prices in a dictionary. Allow the user to enter a product name and display its price if it exists in the dictionary.

- **User Input:** Product name (string) entered by the user.
- **Data Types:** Use dict with string keys and float/int values for prices.

**Q6. Find the Average Marks of Each Student from a CSV File** Write a program to read student names and marks from **week_6_set2_q6.csv** and use a dictionary to calculate and display the average marks of each student.

- **User Input:** No user input required.
- **Data Types:** Use dict with string keys (student names) and list of int values (marks).
- **File Reference:** Refer to **week_6_set2_q6.csv** for input.

**Q7. Remove Blank Lines from a Text File** Write a program to read a text file and write all non-empty lines into another text file, removing blank lines.

- **User Input:** Input file name and output file name entered by the user.

- **Data Types:** Use string for lines.
- **File Handling:** Use text file read mode (r) and write mode (w).

**Q8. Write a Python program that asks the user to enter the names and grades of five students. Store the data in a list of lists and write the results into a CSV file named week_6_set2_q8.csv.**

- **Data Types: string for names, string for grades (A/B/C).**

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